Standard Operating Procedure: Resolving Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node BTM-ACC-34

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8279

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node BTM-ACC-34 occurs due to the degradation of the SFP/XFP transceivers or fiber cuts in the optical transport network. This failure domain is specific to the Access role on Huawei equipment because the Optical Transport Network (OTN) is responsible for transporting client signals over optical fiber. The Huawei Access Node BTM-ACC-34, being a part of this OTN, is susceptible to SFP/XFP degradation or fiber cuts, leading to signal loss and degradation. The Alarm Code ALARM_8279 indicates a minor severity issue, but it is essential to diagnose and resolve the issue promptly to prevent further network degradation.

Pre-requisite Checks

- Verify that the remote hands are dispatched to BTM Layout Aggregation for physical access.

- Check the status of the redundant link to ensure it is operational.

- Perform a visual inspection of the optical fibers and SFP/XFP transceivers for any signs of damage or degradation.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-interfaces, Expected Output: The output will display the status of all optical interfaces, including the SFP/XFP transceivers and their corresponding optical fibers., Description: This command will help identify any issues with the optical interfaces, including SFP/XFP degradation or fiber cuts.

Command: display optical-interfaces optical-power, Expected Output: The output will display the optical power levels of all optical interfaces., Description: This command will help identify any issues
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with the optical power levels, including Rx/Tx power loss.

Command: display optical-interfaces error, Expected Output: The output
will display any errors related to the optical interfaces.,
Description: This command will help identify any issues with the
optical interfaces, including SFP/XFP degradation or fiber cuts.
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Resolution Steps

Step 1: Step: Replace SFP/XFP Transceivers, Command: replace optical-interface Optical-Interface-name, Description: Replace the degraded SFP/XFP transceivers with new ones to restore the optical interface to normal operation.

Step 2: Step: Repair or Replace Fiber, Command: repair optical-fiber Optical-Fiber-name, Description: Repair or replace the damaged or cut fiber to restore the optical transport network to normal operation.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: adjust optical-power Optical-Interface-name, Description: Adjust the Rx/Tx power levels to ensure that the optical interface operates within the normal power range.

Post-Check Verification

```
Command: display optical-interfaces, Expected Output: The output will
display the status of all optical interfaces, including the SFP/XFP
transceivers and their corresponding optical fibers., Description:
Verify that the optical interfaces are restored to normal operation.

Command: display optical-interfaces optical-power, Expected Output:
The output will display the optical power levels of all optical
interfaces., Description: Verify that the optical power levels are
within the normal range.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Transceivers, Command: undo replace optical-interface Optical-Interface-name, Description: Restore the original SFP/XFP transceivers to their previous state in case the replacement fails.

- Step: Restore Original Fiber, Command: undo repair optical-fiber Optical-Fiber-name, Description: Restore the original fiber to its previous state in case the repair or replacement fails.

- Step: Restore Original Rx/Tx Power Levels, Command: undo adjust optical-power Optical-Interface-name, Description: Restore the original Rx/Tx power levels to their previous state in case the adjustment fails.